# Pool ball Recognition Assignment

Computer Vision (CSU44053)

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## 1 Assignment Overview

The objective of this assignment was to develop a computer vision system capable of locating and recognising pool balls from photographs of a pool table. The two main tasks were to determine the position of each ball and to classify it as a solid, striped (along with each ball's repective number) or white cue ball.

The image dataset consisted of one reference image, AllBalls.jpeg, used for parameter tuning, and six test images, Table1–Table6.jpeg, containing different ball arrangements. All images were captured from the same camera viewpoint, and the four corners of the playing surface were provided.

The system was implemented in C++ using OpenCV, following a classical image-processing approach rather than a neural network. The implemented pipelines combines circular Hough detection, geometric filtering, perspective transformation, colour segmentation and rule-based classification.

## 2 System Design and Implementation

### 2.1 Pipeline Overview

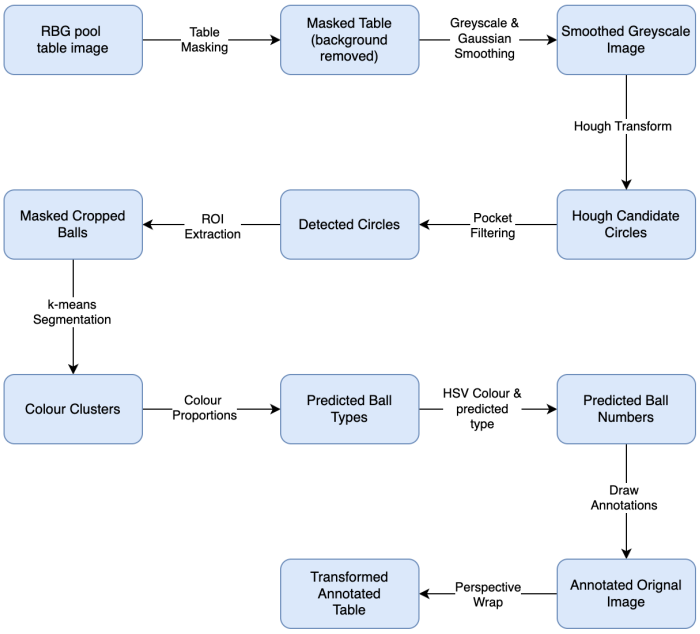



Figure 1: Overall processing pipeline, from input image to detected and classified balls.

The processing pipeline consists of two main stages: ball localisation and ball recognition. Localisation uses geometric and edge information to identify candidate circles. Ball recognition analysed the colour composition of each detected ball.

## 2.2 Ball Detection

Since the table boundaries were known and the camera position remained fixed, the playing surface could be represented by a quadrilateral. A binary mask was constructed from its four corners, restricting circle detection to the relevant image region and reducing interference from the surrounding environment. The goal of this step was to remove the background surrounding the pool table so that it would not interfere with ball detection.

The masked image was converted to grayscale and smoothed using a $5 \times 5$ Gaussian filter with a standard deviation of 1.5 (TODO: explain why these smoothing params). Smoothing was used as it reduces high-frequency noise and small local intensity variations that might otherwise introduce spurious edge responses.

Candidate ball locations were detected using OpenCV's circular Hough transform with the gradient-based method (`HOUGH_GRADIENT`). Applied to the grayscale, Gaussian-smoothed image, it identifies circular structures and returns their estimated centres and radii. These circles were treated as candidates because other features, such as pockets, could also produce detections.

### 2.2.1 Parameter tuning using grid search

The Hough transform requires several parameters that affect how many circles are detected and how accurately they match the balls. Rather than choosing these values only by visual inspection, a grid search was used to compare parameter combinations on the masked tuning image, `AllBalls.jpeg`.

The search varied four parameters:

- Accumulator resolution ratio: $dp \in \{1.0, 1.5, 2.0\}$.
- Minimum distance between detected centres: $\{15, 18, 20, 22, 25\}$ pixels.
- Upper Canny edge threshold: $\{75, 100, 125, 150\}$.
- Circle accumulator threshold: $\{15, 20, 25, 30, 35, 40, 45, 50\}$.

This produced 480 parameter combinations. The minimum and maximum circle radii were fixed at 15 and 30 pixels respectively.

For each combination, detected centres were transformed into table coordinates and matched to the annotated centres within a tolerance of 15 pixels. The best configuration was selected by first maximising the number of true positives, then minimising false positives, and finally minimising the mean centre localisation error.

The results for every combination were saved to a CSV file, together with an image showing the detections from the best configuration. Pocket filtering was not applied during this search; it was evaluated as a separate processing step.

### 2.2.2 Removing pocket detections

Table pockets can also produce circular detections. To remove these, candidate centres were transformed into table coordinates and compared with six approximate pocket positions. Candidates within 25 pixels of a pocket centre were discarded.

After the choosing the final Hough parameters and pocker filterting we obtained the following results on the tuning image:



Figure 2: AllBalls Ball detection with final Hough Params.

## 2.3 Ball Extraction and Colour Segmentation

Once the candidate balls have been detected, each ball is extracted from the original colour image using a square region centred on the predicted circle. A circular binary mask restricts subsequent processing to pixels belonging to the ball.

Initial experiments investigated colour thresholding, including HLS-based methods. However, a single global threshold was difficult to apply consistently across differently coloured balls, particularly in the presence of highlights, shadows and white markings.

The final implementation therefore uses K-means clustering to partition the colour pixels of each ball into two groups. Clustering is performed on BGR pixel values, and the resulting cluster centres are converted to HSV for interpretation.

The proportion of ball pixels assigned to each cluster is calculated. Each cluster's representative colour is then converted to HSV and categorised using the following rules:

- **Light:** brightness $V \geq 170$ and saturation $S \leq 165$.
- **Dark:** brightness $V \leq 90$.
- **Coloured:** all remaining clusters.

Here, saturation and brightness range from 0 to 255. The percentages of clusters in each category are added together to obtain the total light, dark and coloured proportions used for ball-type classification.

This approach reduces the original pixel-level information to a small set of interpretable features. It does not require manually defining a separate colour threshold for every individual ball.

## 2.4 Ball recognition

Ball recognition is performed in two stages. First, each detected ball is classified as white, solid or striped. Its predicted type is then combined with its colour to estimate its number.

### 2.4.1 Ball Type classification

Classification is performed using the proportions of light, dark and coloured pixels obtained from segmentation.

A ball is classified as white when at least 85% of its pixels belong to light clusters. It is classified as solid when at least 70% of its pixels belong to non-light clusters. Balls containing both light and non-light regions, without meeting the preceding criteria, are classified as striped.

These rules reflect the visual characteristics of pool balls: solid balls are predominantly coloured, while striped balls contain a substantial white region in addition to their coloured band.

However, this approach is sensitive to the visible orientation of the ball. A striped ball viewed from its white side may resemble the cue ball, while one viewed predominantly from its coloured side may resemble a solid ball.

An additional experimental classifier was developed using connected-component analysis. It examines the spatial organisation of light and non-light regions, including the largest connected components and

the proportion of the circular boundary occupied by light pixels. Its performance is compared with the simpler colour-based method in Section 3.

#### 2.4.2 Ball number recognition

Number recognition uses the predicted ball type and the HSV colour of the largest eligible non-light cluster. Clusters covering less than 2% of the ball area are ignored, as are light clusters ($V \geq 170$ and $S \leq 165$).

A ball predicted as white is assigned number 0, representing the cue ball. For solid and striped balls, the selected cluster is examined using the following rules. OpenCV represents hue on a scale from 0 to 179, while saturation and brightness range from 0 to 255.

If $S < 60$, the ball is assigned number 8 only when $V \leq 90$ and its predicted type is solid. Otherwise, this low-saturation case returns an unknown result.

For clusters with $S \geq 60$, the colour thresholds in Table 1 determine the base number.

Table 1: Experimental HSV thresholds for ball-number recognition.

| Colour | Base number | HSV condition |
|---|---|---|
| Yellow | 1 | $15 \leq H < 35$ |
| Blue | 2 | $100 \leq H < 130$ |
| Red | 3 | ($H < 6$ or $H \geq 170$), $V \geq 150$ |
| Purple | 4 | $130 \leq H < 165$ |
| Orange | 5 | $6 \leq H < 15$ |
| Green | 6 | $45 \leq H < 90$ |
| Maroon | 7 | ($H < 6$ or $H \geq 170$), $V < 150$ |

Solid balls retain the base number, while striped balls add 8, giving numbers 9–15. Red and maroon share the same hue range, so brightness is used to distinguish them.

When segmentation is unavailable, the predicted type is unknown, or no eligible cluster or colour match is found, the predicted number is set to $-1$. These thresholds are experimental and may be sensitive to changes in lighting.

### 2.5 Perspective Transformation

The table appears at an angle in the original photographs, while the provided ground-truth positions refer to a rectangular, top-down view. To compare our detections with these positions, their coordinates must be converted into the same coordinate system.

The four known table corners were mapped to the corners of a $1434 \times 683$ pixel rectangle. The resulting perspective transformation was then applied to each detected ball centre.

Ball detection and recognition were performed on the original image. Transforming the image beforehand could stretch the circular ball outlines into ellipses, making the Hough circle detector less reliable. The full image was transformed separately when producing visualisations for comparison with the ground truth.

TODO: Add image of the AllBalls image after perpective transformation

## 3 Evaluation and Results

### 3.1 Methodology

The reference image AllBalls.jpeg was used exclusively for parameter selection, while the six other images were reserved for evaluation. This separation reduces the risk of reporting performance only on an image used to optimise the detector.

An automated grid search evaluated 480 combinations of Hough parameters. Configurations were ranked by the number of correctly detected balls, followed by the number of false positives and the average localisation error.

The final parameters were:

Table 2: Final Hough transform parameters for ball detection.

| Parameter | Value | Description |
|---|---|---|
| `dp` | 1.5 | Image-to-accumulator resolution ratio |
| `minDist` | 18 | Minimum distance between centres (pixels) |
| `param1` | 75 | Upper Canny edge threshold |
| `param2` | 35 | Accumulator threshold for accepting circles |
| Minimum radius | 15 | Smallest accepted circle radius (pixels) |
| Maximum radius | 30 | Largest accepted circle radius (pixels) |

Detected centres were compared with the provided ground truth after perspective transformation. A one-to-one matching procedure assigned detections to annotated centres within a maximum distance of 15 pixels in table coordinates.

Unmatched detections were counted as false positives (FP), while unmatched ground-truth balls were counted as false negatives (FN). Correctly matched detections were counted as true positives (TP).

Performance was assessed using precision, recall and F1-score. The mean Euclidean distance between matched centres was also calculated to evaluate localisation accuracy.

## 3.2 Ball Detection Results

On the tuning image, the selected configuration detected all 16 annotated balls, with three additional false detections. This gave a precision of 84.21%, recall of 100% and F1-score of 91.43%, before pocket filtering.

The subsequent evaluation on the six test images produced the following results using the full masked and pocket-filtered detection pipeline.

Table 3: Ball detection results on the validation images.

| Image | GT | TP | FP | FN | F1 |
|---|---|---|---|---|---|
| Table 1 | 16 | 14 | 2 | 2 | 0.8750 |
| Table 2 | 16 | 16 | 2 | 0 | 0.9412 |
| Table 3 | 16 | 15 | 0 | 1 | 0.9677 |
| Table 4 | 15 | 14 | 1 | 1 | 0.9333 |
| Table 5 | 14 | 14 | 1 | 0 | 0.9655 |
| Table 6 | 13 | 13 | 1 | 0 | 0.9630 |
| **Total** | **90** | **86** | **7** | **4** | **0.9399** |

Across all six images, the detector achieved an overall precision of 92.47%, recall of 95.56% and F1-score of 93.99%. The mean localisation error across matched detections was approximately 5.01 pixels.

The relatively high recall indicates that the method successfully identifies most of the visible balls. However, its performance is less consistent when balls overlap or are closely grouped.

Table 1 produced the lowest detection F1-score and the largest mean centre error (7.68 pixels). This image contains a tightly arranged group of balls, where adjacent circular boundaries may interfere with Hough detection and accurate localisation.

By contrast, the later images contain more spatially separated balls, making individual circular outlines easier to identify. This is consistent with the generally stronger detection results on Tables 3–6.

## 3.3 Ball Recognition Results

Recognition was evaluated separately using only the detections successfully matched to ground-truth balls. This isolates classification performance from localisation errors.

The baseline K-means colour classifier correctly classified 76 of the 86 matched balls, giving an accuracy of 88.37%.

The experimental classifier incorporating connected-component features correctly classified 73 of the 86 matched balls, reducing accuracy to 84.88%.

Table 4: Ball recognition results for two different methods.

| Method | Correct/Matched | Accuracy | | | |
|---|---|---|---|---|---|
| K-means colour rules | 76/86 | 88.37% | | | |
| K-means + CCA + colour rules | 73/86 | 88.37% | | | |
| **Total** | **90** | **86** | **7** | **4** | **0.9399** |

The additional connected-component rules introduced three new classification errors without correcting any existing errors in this test set.

This suggests that spatial information is not necessarily more reliable than overall colour proportions. In particular, the shape and boundary coverage of white regions vary considerably with the orientation of striped balls. Consequently, rules intended to recognise stripes can also incorrectly modify predictions for the white cue ball or solid balls.

The simpler colour-based method was therefore retained in the main application.

The recognition accuracy of 88.37% is conditional on successful detection. Considering all 90 ground-truth balls, the system correctly detected and classified 76, corresponding to an end-to-end correct identification rate of 84.44%.

This distinction is important because a classification system cannot correctly identify a ball that was missed during localisation.

## 3.4 Qualitative Analysis and Failure Cases

Visual inspections revealed several recurring challenges.

**Overlapping and touching balls**. When balls are closely arranged, their visible boundaries may merge or be partially obscured. This can result in missed circles, imprecise centres or incorrect segmentation caused by pixels from neighbouring balls.

**Ball orientation**. A striped ball does not necessarily display equal amounts of white and colour. Depending on its orientation, the observed colour proportions may resemble either a solid ball or the white cue ball. This limitation is inherent to classification based on a single two-dimensional view.

**Lighting and shadows**. Highlights may cause coloured areas to appear white, while shadows can make coloured regions appear dark. Because classification depends on representative cluster colours and fixed HSV thresholds, these variations can change the predicted ball type.

**Detection errors propagated into recognition**. An inaccurate circle may include background pixels or exclude part of the ball. K-means then segments an imperfect crop, and the resulting percentages may no longer represent the actual ball.

These observations demonstrate why good localisation performance does not guarantee equally strong recognition performance.

# 4 Solution Anaysis and limitations

## 4.1 Limitations

Our method performs well on the supplied dataset, but several design choices limit its generalisation.

First, the fixed table mask and transformation depend on a known camera viewpoint. A different viewpoint would require new corner coordinates or an automatic table-localisation stage.

Second, a single set of Hough parameters may not be suitable for images containing balls with substantially different apparent sizes, lighting conditions or resolutions.

Third, two-cluster colour segmentation simplifies classification but may not adequately represent balls containing several visually distinct regions, such as the coloured band, white area, printed number and specular highlights.

## 4.2 Possible Improvements

Potential improvements include more robust ball-boundary extraction, adaptive colour classification and improved treatment of overlapping balls. More extensive testing across varied lighting conditions and viewpoints would also help assess generalisation.

Nevertheless, these extensions are not essential to the current assignment. The existing pipeline provides a complete solution with measurable performance and well-understood limitations.

# 5 Conclusion

This project implemented a classical computer vision pipeline for pool-ball localisation and recognition using geometric information, circular Hough detection, perspective transformation and colour-based segmentation.

The detector achieved an overall F1-score of 93.99% on the six test images, while the selected colour-based recognition method correctly classified 88.37% of successfully matched balls.

The results demonstrate that computer vision methods can produce strong localisation performance when the camera geometry is known. Recognition is more challenging because the visible colour distribution changes with ball orientation, illumination and occlusion.

The comparison with connected-component analysis also demonstrated that additional processing does not necessarily improve accuracy. Overall, the project achieved its two primary objectives while identifying concrete limitations and directions for future work.

# 6 Appendix

## 6.1 Use of Generative AI

## 6.2 Resources